Examination of magnetic properties by measuring hysteresis curves of different materials

J. Bergmeier and S. Steinicke

Fe-Ni 5000 H2 was found to have a remanence of $B_{r,5000H2} = (0.655 \pm 0.021)$T, a coercivity of $H_{c,5000H2} = (3.41 \pm 0.35)\frac{\mathrm{A}}{\mathrm{m}}$, and a saturation induction of $B_{s,5000H2} = (1.4660 \pm 0.0071)$T. For Fe-Ni 5000 Z the values $B_{r,5000Z} = (1.4295 \pm 0.0071)$T, $H_{c,5000Z} = (12.80 \pm 0.35)\frac{\mathrm{A}}{\mathrm{m}}$, and $B_{s,5000Z} = (1.4435 \pm 0.0071)$T were obtained. And for steel St37K we measured $B_{r,steel} = (0.551 \pm 0.021)$T and $H_{c,steel} = (403 \pm 28)\frac{\mathrm{A}}{\mathrm{m}}$. We calculated $\mu_{r,wood}$ to be 0.91052 ± 0.00042. The density of energy lost in one revolution of the hysteresis curve was determined to be $\rho_{E,5000H2} = 0.002607164\,\frac{\mathrm{J}}{\mathrm{kg}}$ and $\rho_{E,steel} = 0.28080897\,\frac{\mathrm{J}}{\mathrm{kg}}$.

Execution of experiment: April 1, 2022
Submission: April 6, 2022

1 Introduction

Magnets and magnetic properties are used in a multitude of technologies, such as HDD data storage media, microphones and speakers, MRI medical examinations, and electromagnets for moving large pieces of metal such as cars. The behavior of objects, when brought into a magnetic field, is dependent on the type of material. To visualize this behavior, a hysteresis curve can be drawn. It shows the magnetic field B as a function of the H-field. This connection is not technically a function because the value measured does not just depend on H but also on the magnetic history of the material. As examples of different behavior Ni-Fe 5000 H2, wood, steel St37K, and Fe-Ni 5000 Z were chosen. Among other things, remanence, coercivity, and the maximum B-field were measured.

2 Theory

In this paper, the magnetic properties of ferromagnetic materials are studied to a large extent which is why we first want to shed light on the theory behind them. Ferromagnetic materials are characterized by the so-called *magnetic domains*. This means that the magnetic phenomena are no longer to be understood on an atomic level but on a microcrystalline level. Within these domains, the magnetic electron moments and spins are aligned identically due to exchange forces. Without an external field, the statistical alignment of the domains results in an average magnetic effect of 0. The limits of these areas are called *domain walls*.
When a magnetic field H is increased, the domain walls shift until all spins are aligned parallel to the field. The necessary magnetization is called *saturation magnetization*. From here on, the magnetic flux B increases with the vacuum permeability μ_0 as its slope, while previously there was a nontrivial function *B(H)*. The magnetization M is also included here, due to the alignment of the magnetic moments.

$$B = \mu_0(M + H) \tag{1}$$

Not only the alignment of the spins plays a role in the hysteresis curve, but also the thermal fluctuation that counteracts it.
Because of this effect, when the magnetic field is reduced, some *magnetic domains* become statistically aligned again, causing the flux density to decrease. However, the spin-spin coupling is larger, and the $B(H)$ curve is split. This leaves a residual magnetization, the so-called *remanence* B_r, even if there is no external field.
The magnetic field which is necessary to make the flux disappear is called *coercivity field* H_c. By increasing the external field with opposite polarity, another saturation will be reached. Therefore, the hysteresis curve is identically for both polarities of the field and is always run counterclockwise.
Lastly, by reversing the polarity and minimally reducing the magnetic field, it is possible to demagnetize the

material since a spiral around the zero point will be created. This effect will be examined in section 5. [2]

3 Hysteresis curve of Fe-Ni-eutectics (5000 H2)

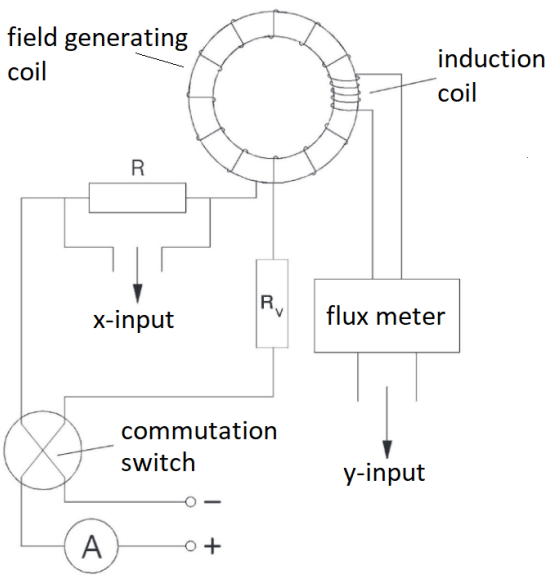

Figure 1: Experimental setup: Depending on the voltage drop at the resistor R or Rv, the induced voltage drop at the fluxmeter is recorded on the Pasco interface. This is generated by the field-generating coil. The commutation switch is used to reverse the polarity of the current so that a hysteresis curve can be measured. [1]

The experiment is set up as shown in fig. 1. A big coil is rolled around a solid donut-shaped core of differing materials. This inductivity is connected to a power source with variable polarity which is controlled by the commutation switch. In the same circuit, there are two resistors that can be used to measure the current flowing through the system via the voltage drop U along the resistors R or R_v. For measurements with a high current, the built-in resistor R is used. For small currents, the voltage drop across the series resistor R_v is recorded. A second coil is wrapped around the same core and hence is inductively coupled with the first coil. The fluxmeter is connected to the induction coil and measures a voltage spike which is identical to the change in magnetic flux $N_2 \cdot \Phi$ inside the coil. To determine the hysteresis curve $B(H)$, the quantities magnetic field H and magnetic flux density B are calculated as shown below.[1]

$$H = \frac{N_1}{\pi \frac{1}{2}(d_i + d_o)R} U \tag{2}$$

$$B = \frac{N_2 \Phi}{N_2 A} \tag{3}$$

$$A = \frac{1}{2}(d_o - d_i) \cdot h \cdot f \tag{4}$$

The area of the coil's core is determent by the outer and inner diameters d_o and d_i, the height h, and the filling factor f. Furthermore, the number of windings N_1 and N_2 are specified by the manufacturer.
Resistor R_v was measured with the *Voltcraft VC170*, the instruments error was taken from the handbook[3].

$$R_v = (66.8 \pm 1.4)\Omega$$

The maximum current was set to $I_{max} = 100\,\mathrm{mA}$ and the voltage was increased until this amperage was hit. During the recording of the hysteresis curve, the voltage was lowered to zero, then, with changed direction, it was increased again until I_{max} was reached. This procedure was then reversed. To make sure the end of the curve overlaps with its start, the fluxmeter was zeroed at the beginning of each measurement. All the hysteresis curves in this paper were recorded analogous.

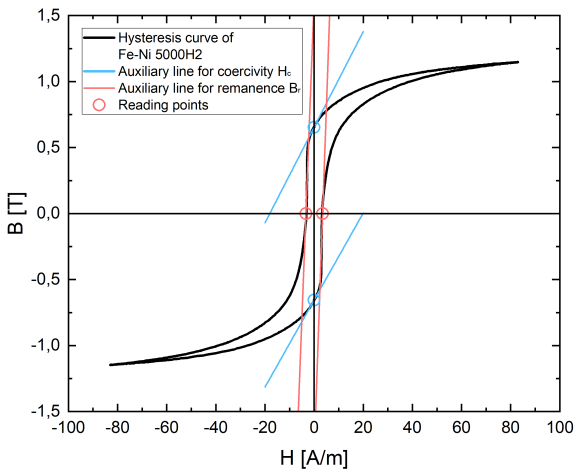

Figure 2: Hysteresis curve for Fe-Ni alloy 5000 H2 with circles marking where the remanence (blue line) and coercivity (red line) were read off. The maximum current through the circuit was 0.1 A.

Fig. 2 shows the hysteresis for core *Fe-Ni 5000 H2*. Because zero, as measured by the flux meter, did drift during the measurement, we first tried to correct for this deviation. In the formula eq. 5 used for correction, it was assumed that the drift is linear with time.

$$\frac{\mathrm{d}B}{\mathrm{d}t} = \frac{B_{end} - B_{start}}{t_{end} - t_{start}} \tag{5}$$

$B_{start/end}$ are the averages of the magnetic field measured for the same H once at the beginning and later at the end of the observation period. $t_{start/end}$ are the corresponding averaged times. The curve was also moved in the x/y-directions to bring the middle point to the coordinate origin. For this offset correction, it was assumed that the curves are point-symmetric. These corrections were made for all graphs and all calculated values in this paper.
Remanence B_r (blue line) and coercivity H_c (red line) were read from fig. 2:

$$B_{r,5000H2} = (0.655 \pm 0.021)\mathrm{T}$$
$$H_{c,5000H2} = (3.41 \pm 0.35)\frac{\mathrm{A}}{\mathrm{m}}$$

The errors were determined graphically.

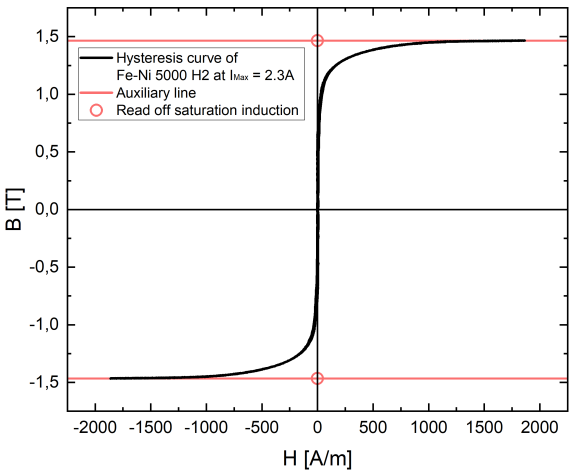

Figure 3: Shown is the hysteresis curve for Fe-Ni 5000 H2 with the current limited to 2.3 A. The points, where the saturation induction was read off, are marked with red circles.

To determine saturation induction, I_{max} was increased to 2.3 A and the resistor R_V was removed. The curve that can now be observed is plotted in fig. 3. Through the horizontal parts of the hysteresis the red lines were drawn. Averaging their B-locations gives the saturation induction B_s, the error is calculated from the graphical accuracy and gaussian propagation of uncertainty.

$$B_{s,5000H2} = (1.4660 \pm 0.0071)\mathrm{T}$$

The alloy Permenorm 5000 H2 consists of approx. 50 % nickel and 50 % iron. Compared to the literature value $H_c = 3.5\,\mathrm{A/m}$ [4], the measured one is within the error bar identical. This does not apply to the saturation induction with a literature value of $B_s = 1.55\,\mathrm{T}$ [5]. The reason for this is with high probability that on the one hand, the error was estimated too small and therefore possible systematic error sources are not included and on the other hand, the nickel-iron distribution is not exactly the same. The low coercivity and high saturation induction characterize this class of materials and are therefore often used to measure currents or to conduct magnetic flux. Examples of applications are: Measuring transducers, Fi-switches, or coil cores for stepper motors in quartz clocks [6].

4 Measurements with wooden core

The same measurement from chapter 3 is repeated with a coil wined around a wooden core. For comparison, the hysteresis curve was recorded in the same 100 mVs resolution that worked well for *Fe-Ni 5000 H2* and also in the much higher 2 mVs resolution of the fluxmeter. The high-resolution curve is shown in fig. 4. At the fluxmeter's lowest sensitivity only a horizontal line at 0 V was discernable. Because the voltmeter is at its limit no further conclusions can be drawn from this. Wood does

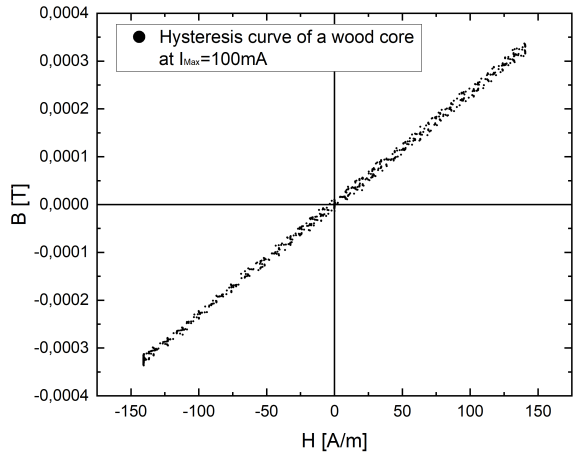

Figure 4: The function B(H) was plotted for a coil with a wooden core. The current was limited to 0.1 A.

not interact strongly with magnetic fields. Its magnetic permeability $\mu_{r,wood}$ is expected to be close to the value of vacuum ($\mu_{r,vacuum} = 1$). If the resolution of the fluxmeter is increased significantly, it can be seen that the line has a non-zero slope s which differs from vacuum. Because of the high resolution, the drift of the fluxmeter's zero was very relevant, and cleaning it up made a clear difference. The actual value can be obtained through eq. (6) with the slope which was determined with error via a linear fit and the vacuum permeability which has the value $\mu_0 = 4\pi \cdot 10^{-7}\,\frac{\mathrm{H}}{\mathrm{m}}$.

$$s = \mu_0 \mu_{r,wood} \tag{6}$$
$$\mu_{r,wood} = 0.91052 \pm 0.00042$$

It can be seen that $\mu_{r,wood}$ does not differ much from vacuum. This is the reason why even strong H-fields do not seem to induce a B-field inside the wooden core if

we compare it to metal alloys. With $\mu_{r,wood} \leq 1$, wood is classified as a diamagnetic material which means that this behavior will not change with temperature. Without an external magnetic field, diamagnetic materials have no magnetic field of their own, they are non-magnetic.

5 Hysteresis curve of steel core

The hysteresis curve fig. 5 was measured on *flat steel St37K* with the power supply limited to 5 A. Again, the

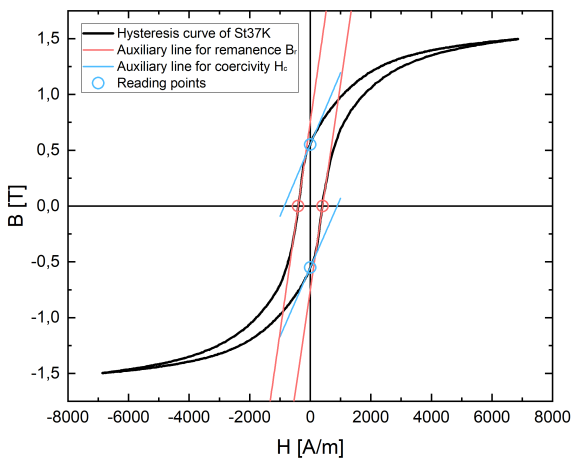

Figure 5: Measurement of the magnetic induction as a function of the H-field for steel St37K. The remanence is marked in blue, the coercivity in red.

remanence $B_{r,steel}$ (blue line) and coercivity $H_{c,steel}$ (red line) were graphically determined.

$$B_{r,steel} = (0.551 \pm 0.021)\mathrm{T}$$
$$H_{c,steeel} = (403 \pm 28)\frac{\mathrm{A}}{\mathrm{m}}$$

To show that $B(H)$ cannot be reversed a curve with the same settings used in fig. 5 was recorded, but with the difference that the voltage was only decreased until the current was approximately 0.5 A. The voltage was then increased a bit, before continuing with the curve in the normal fashion. On the return journey, we proceeded equivalently. The course of the hysteresis can be seen in fig. 6. It can be observed that on a direction change the induced magnetic field does not follow the same path back, but rather takes a loop outside the already measured range. This is the case because the history of the material is relevant for the observed inductivity. The spins of the metal atoms get oriented along one axis by the outside magnetic field. If the field is decreased, they slowly revert to chaos. This means that by the second measurement of the same H value the spins inside the metal were oriented much more randomly and hence were not able to create a magnetic field as strong as before. With the knowledge that the field is less strong if an H-value is approached from lower values, the direction of rotation inside the loop can be worked out. This dependency on the material's history is the reason why in this paper the curves observed were not usually functions, but rather hysteresis curves with different B-values for the same H. Furthermore, the orbital direction of a hysteresis curve is always counterclockwise.

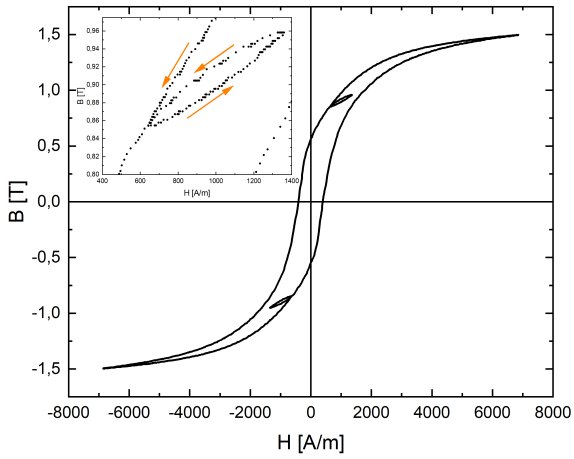

Figure 6: Hysteresis curve of St37K, but at currents of 0.5 A the direction of the revolution was reversed for a short while. The second, smaller graphic shows the picture zoomed in on the point where reversion took place. Orange arrows mark the direction of revolution inside the loop.

How metal can be demagnetized is shown in fig. 7. According to the principle explained in the paragraph above the voltage is raised to progressively lower values on each revolution, until there is no remanence left at $H = 0$. With this method, we achieved complete statistical distribution of the spin orientations inside the metal.

6 Hysteresis curve of Fe-Ni-alloy 5000Z

Once again, a hysteresis curve is measured. This time for $5000Z$, a different Fe-Ni-alloy. The curve is shown in fig. 8 and the remanence, coercivity, and saturation induction are calculated in the same way as previously.

$$B_{r,5000Z} = (1.4295 \pm 0.0071)\mathrm{T}$$
$$H_{c,5000Z} = (12.80 \pm 0.35)\frac{\mathrm{A}}{\mathrm{m}}$$
$$B_{s,5000Z} = (1.4435 \pm 0.0071)\mathrm{T}$$

A comparison with the literature values $B_s = 1.65\,\mathrm{T}$ and $H_c = 0.12\,\mathrm{A/cm}$ shows that the measured quan-

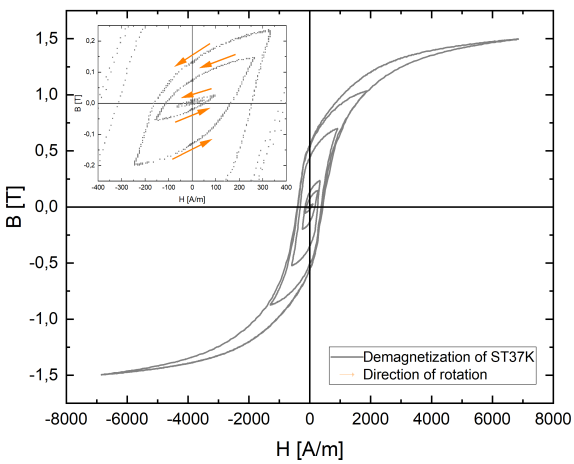

Figure 7: The induction inside steel St37K during the process of demagnetization. A second, smaller graphic shows the view zoomed into the middle of the spiral with the direction of measurement marked by orange arrows.

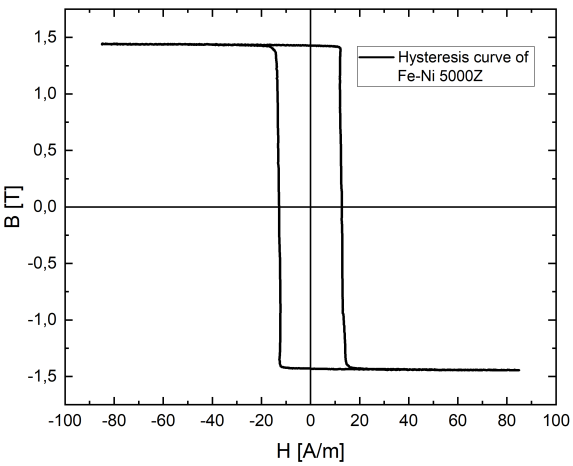

Figure 8: Hysteresis curve of iron-nickel alloy 5000 Z.

tities within the errors do not match them. However, the difference is only very small and due to the small error estimate.

7 Comparison of energy-loss

Every time one lap around a hysteresis curve is completed, a portion of the expended energy is lost. This loss is equal to the area inside the curve. The energy lost per unit of mass can be calculated from the density ρ of the material with eq. (7).

$$\rho_E = \frac{1}{\rho} \cdot \oint B(H)\, dH \tag{7}$$

This inefficiency arises during the reorientation of the spins and is dissipated in form of heat. The energy density ρ_E was calculated for Fe-Ni 5000 H2 and steel with the densities provided in the instructions [1]. Errors are not provided because we cannot estimate to what extent the errors of the voltage measurement cancel out when the curve as a whole is considered.

$$\rho_{E,5000H2} = 0.002607164 \frac{\mathrm{J}}{\mathrm{kg}}$$
$$\rho_{E,steel} = 0.28080897 \frac{\mathrm{J}}{\mathrm{kg}}$$

If the curves of wood, Fe-Ni 5000Z/H2, and steel are compared, it can be seen that Fe-Ni 5000 Z would be considered a magnetically hard material, while all other investigated curves belong to magnetically soft materials. A soft magnet has a narrow curve which encloses only a small area. Their energy losses per revolution are small and hence these materials are used in electric motors, generators, and transformers. Hard magnetic materials have almost square hysteresis which maximises the enclosed area. They are particularly useful as electromagnets and inside devices to store information. Because of the different waste energy densities, steel is much less suitable for a lot of technical applications compared to Fe-Ni 5000 H2 even though they are both considered soft materials.

8 Summary

In this paper hysteresis curves were measured for the materials Fe-Ni (5000 H2 & 5000 Z), steel (St37K), and wood. The remanences: $B_{r,5000H2} = (0.655 \pm 0.021)$T, $B_{r,5000Z} = (1.4295 \pm 0.0071)$T, and $B_{r,steel} = (0.551 \pm 0.021)$T; the coercivities: $H_{c,5000H2} = (3.41 \pm 0.35)\frac{\mathrm{A}}{\mathrm{m}}$, $H_{c,5000Z} = (12.80 \pm 0.35)\frac{\mathrm{A}}{\mathrm{m}}$, and $H_{c,steel} = (403 \pm 28)\frac{\mathrm{A}}{\mathrm{m}}$; and the saturation inductions: $B_{s,5000H2} = (1.4660 \pm 0.0071)$T and $B_{s,5000Z} = (1.4435 \pm 0.0071)$T were determined graphically. The permeability of wood was calculated to be $\mu_{r,wood} = 0.91052 \pm 0.00042$ which we found to be close to that of vacuum, which means

wood does not interact with magnetic fields in a strong way. The density of energy lost in one revolution of the hysteresis curve was calculated for steel and one of the Fe-Ni alloys: $\rho_{E,5000H2} = 0.002607164\,\frac{\mathrm{J}}{\mathrm{kg}}$ and $\rho_{E,steel} = 0.28080897\,\frac{\mathrm{J}}{\mathrm{kg}}$. We found Ni-Fe 5000 Z to be a magnetically hard material, while 5000 H2, wood, and steel St37K are considered soft.

References

[1] Physikalisches Grundpraktikum, Universität Würzburg, Modul C2, Versuch 39, "Wärmestrahlung", 2022

[2] Frauenfelder, Huber, Einführung in die Physik, Band II, Ernst Reinhard Verlag, Basel, 1958

[3] Voltcraft, Digital Multimeter Operating Instructions, Version 02/11

[4] "Patents Weichmagnetische nickel-eisen-legierung mit kleiner koerzitivfeldstärke, hoher permeabilität und verbesserter korrosionsbeständigkeit ", https://patents.google.com/patent/WO1999039358A1, last visited: April 5, 2022.

[5] magnetic properties of Ni-Fe alloys, https://vacuumschmelze.de/Produkte/Weichmagnetische-Werkstoffe-und-Stanzteile/40-bis-50-Nickel-Eisen, last visited: April 5, 2022.

[6] Examples of applications of 5000H2, https://docplayer.org/20879624-Weichmagnetische-werkstoffe-und-halbzeuge.html, last visited: April 5, 2022.

[7] Michalowsky, Schneider, Magnettechnik, 3.Auflage, Vulkan-Verlag